Calibrated Goodness-of-Fit Test for Effective-Spin Population Models

Automated Pipeline Report

March 27, 2026

Abstract

We apply a calibrated χ^2 goodness-of-fit test to five population models for the effective inspiral spin parameter χ_{eff} of binary black-hole mergers observed during the LIGO–Virgo–KAGRA O3 and O4 observing runs. The test compares binned distributions of the 5th- and 95th-percentile χ_{eff} posterior quantiles for $N_{\mathrm{ev}} = 71$ events against predictions from re-weighted injection sets. p-values are obtained by Monte Carlo calibration under each model's own null hypothesis. We report results for the Isotropic, Azzalini skew-normal, LVK Default, Truncated Gaussian, and Roulet+ 2021 three-component mixture models.

1 Method

1.1 Observable and binning

For each event i in the catalog we extract the 5th and 95th percentiles of the marginal posterior on χ_{eff}, denoted q_i^5 and q_i^{95}. We form a concatenated histogram vector

$$\mathbf{h} = \left(h_1^5, \ldots, h_{N_{\mathrm{bin}}}^5,\ h_1^{95}, \ldots, h_{N_{\mathrm{bin}}}^{95} \right), \tag{1}$$

where $N_{\mathrm{bin}} = 15$ equally spaced bins cover the range of observed quantiles (with 10% margin on each side). Each sub-histogram is normalised to sum to unity so that $\mathbf{h}$ has $2N_{\mathrm{bin}} = 30$ entries.

1.2 Population models and injection re-weighting

We consider five population models for the χ_{eff} distribution:

Model 1: Isotropic spins. Component spins are uniform in magnitude on $[0, \chi_{\mathrm{max}}]$ and isotropic in orientation. No free parameters; the injection prior itself is the population.

Model 2: Azzalini skew-normal (scipy). The standard Azzalini skew-normal distribution (?), as implemented in `scipy.stats.skewnorm`:

$$p(\chi_{\mathrm{eff}} \mid a, \xi, \omega) = \frac{2}{\omega}\, \phi\!\left(\frac{\chi_{\mathrm{eff}} - \xi}{\omega}\right)\, \Phi\!\left(a\, \frac{\chi_{\mathrm{eff}} - \xi}{\omega}\right), \tag{2}$$

truncated to $[-1, 1]$, where ϕ and Φ are the standard normal PDF and CDF, a is the shape (skewness), ξ the location, and ω the scale. Best-fit: $(a, \xi, \omega) = (13.44, -0.069, 0.161)$. *Note:* This is **not** the LVK epsilon-skew-normal (split-normal) parameterisation.

Model 3: LVK Default. Spin magnitudes follow truncated normals on $[0, 1]$:

$$p(\chi_i \mid \mu_\chi, \sigma_\chi) = \mathcal{N}_T(\chi_i;\, \mu_\chi, \sigma_\chi,\, 0, 1), \qquad i \in \{1, 2\}. \tag{3}$$

Tilt angles are a mixture of an aligned and an isotropic component:

$$p(\cos\theta_1, \cos\theta_2 \mid \mu_t, \sigma_t, \zeta) = \zeta \prod_{i=1}^{2} \mathcal{N}_T(\cos\theta_i;\, \mu_t, \sigma_t,\, -1, 1) + \frac{1-\zeta}{4}. \tag{4}$$

Five free parameters: $(\mu_\chi, \sigma_\chi, \mu_t, \sigma_t, \zeta)$. MAP from GWTC-4.0: $(0.091, 0.318, 0.752, 0.884, 0.382)$.

Model 4: Truncated Gaussian.

$$p(\chi_{\rm eff} \mid \mu, \sigma) = \mathcal{N}_T(\chi_{\rm eff};\, \mu, \sigma,\, -1, 1). \tag{5}$$

Median from GWTC-4.0: $(\mu, \sigma) = (0.042, 0.103)$.

Model 5: Three-component mixture (?, Roulet+ 2021). Three half-Gaussians centered at $\chi_{\rm eff} = 0$:

$$p(\chi_{\rm eff} \mid \zeta_+, \zeta_-, \sigma) = \zeta_0\, \mathcal{N}_T(\chi_{\rm eff};\, 0, \sigma_0,\, -1, 1) + \zeta_-\, \mathcal{N}_T(\chi_{\rm eff};\, 0, \sigma,\, -1, 0) + \zeta_+\, \mathcal{N}_T(\chi_{\rm eff};\, 0, \sigma,\, 0, 1), \tag{6}$$

where $\zeta_0 = 1 - \zeta_+ - \zeta_-$ and $\sigma_0 = 0.04$ (fixed). The first component is a narrow "nonspinning" peak; the second and third are anti-aligned and aligned half-Gaussians with shared width σ. MLE from O1–O3a: $(\zeta_+, \zeta_-, \sigma) = (0.45, 0.00, 0.23)$. The data are consistent with no anti-aligned component ($\zeta_- = 0$).

Figure 1 shows the intrinsic $\chi_{\rm eff}$ probability density for each model at its best-fit hyperparameters, without selection effects.

Each model defines per-injection importance weights $w_k(\boldsymbol{\theta})$ that re-weight the found-injection set from its astrophysical prior to the target population. The predicted histogram $\mathbf{h}^{\rm pred}$ is obtained by building the same binned representation from the weighted injections.

1.3 χ^2 test statistic

1.4 χ^2 test statistic

For a given set of hyperparameters $\boldsymbol{\theta}$, each injection k receives an importance weight $w_k(\boldsymbol{\theta}) = p_{\rm pop}(\chi_{{\rm eff},k}; \boldsymbol{\theta}) \,/\, p_{\rm prior}(\chi_{{\rm eff},k}, q_k)$, where $p_{\rm prior}$ is the isotropic-spin prior of the injection campaign (evaluated analytically following Callister et al.). To estimate the predicted histogram $\boldsymbol{\mu}(\boldsymbol{\theta})$ and its covariance $\hat{\Sigma}(\boldsymbol{\theta})$, we draw $B = 1000$ bootstrap catalogues of $N_{\rm ev}$ events from the injection set with sampling probabilities $\propto w_k(\boldsymbol{\theta})$, histogram each, and compute the sample mean and covariance of the resulting histogram vectors. Bins with zero bootstrap variance (never populated) are projected out before inversion.

The test statistic is then

$$\chi^2(\boldsymbol{\theta}) = \big(\mathbf{h}^{\rm obs} - \boldsymbol{\mu}(\boldsymbol{\theta})\big)^T\, \hat{\Sigma}(\boldsymbol{\theta})^+ \,\big(\mathbf{h}^{\rm obs} - \boldsymbol{\mu}(\boldsymbol{\theta})\big), \tag{7}$$

where $\hat{\Sigma}^+$ denotes the Moore–Penrose pseudoinverse.

1.5 Calibrated p-values

Each population model comes with a set of best-fit hyperparameters $\boldsymbol{\theta}_{\rm BF}$ obtained from hierarchical Bayesian inference on the same event catalogue (see Table 1 for sources). We wish to test: *is $\boldsymbol{\theta}_{\rm BF}$ consistent with the observed catalogue?*

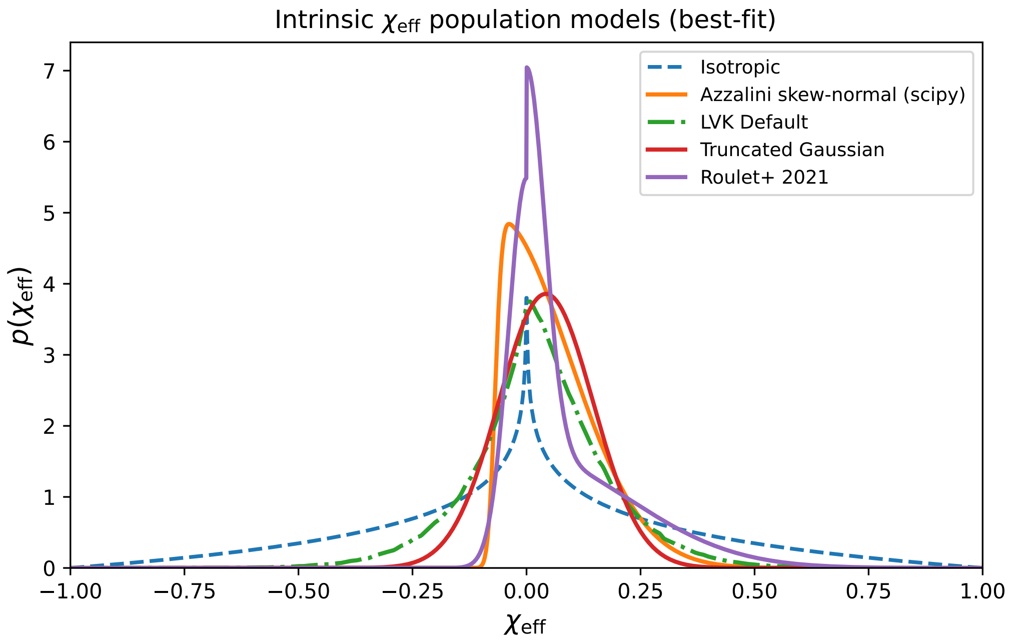

Figure 1: Intrinsic $\chi_{\rm eff}$ population distributions at the best-fit hyperparameters for each model (no selection effects applied). The isotropic model is broad and symmetric, while the skew-normal, truncated Gaussian, and two-Gaussian mixture all peak sharply near $\chi_{\rm eff} \approx 0.1$. The LVK Default model, which operates in component-spin space, produces a broader $\chi_{\rm eff}$ distribution peaked near zero.

A complication is that $\boldsymbol{\theta}_{\rm BF}$ was itself inferred from the data. To account for this, we allow the test statistic to be minimised over $\boldsymbol{\theta}$:

$$\chi^2_{\rm min} = \min_{\boldsymbol{\theta}} \ \chi^2(\boldsymbol{\theta}), \tag{8}$$

using the Nelder–Mead simplex algorithm (3 random restarts from $\boldsymbol{\theta}_{\rm BF}$ and two random points in the prior volume), yielding the best-fit parameters $\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \chi^2(\boldsymbol{\theta})$. For models with no free parameters (Isotropic), $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_{\rm BF}$.

To calibrate the p-value we ask: if the true population were $\boldsymbol{\theta}_{\rm BF}$, how often would a catalogue of $N_{\rm ev}$ events yield a $\chi^2_{\rm min}$ as large as observed?

1. Generate mock catalogue j by drawing $N_{\rm ev}$ events from the injection set with probabilities $\propto w_k(\boldsymbol{\theta}_{\rm BF})$.

2. Beings in mock universe j do not know $\boldsymbol{\theta}_{\rm BF}$. They minimise $\chi^2(\boldsymbol{\theta})$ on their own catalogue, obtaining $\chi^2_{{\rm min},j}$.

3. Repeat for $j = 1, \ldots, N_{\rm mock}$.

The calibrated p-value is

$$p = \frac{1}{N_{\rm mock}} \sum_{j=1}^{N_{\rm mock}} \mathbb{1}\left[\chi^2_{{\rm min},j} \geq \chi^2_{\rm min,\,obs}\right]. \tag{9}$$

This construction accounts for the "look-elsewhere effect" of fitting: because each mock catalogue is also fitted, the null distribution of $\chi^2_{\rm min}$ is shifted to lower values relative to the fixed-$\boldsymbol{\theta}$ distribution, and the resulting p-value is appropriately conservative.

2 Results

Table 1 summarises the goodness-of-fit results for each model. Figures 2–?? show the p-value comparison, example histogram overlay, and null distribution, respectively.

Table 1: Calibrated goodness-of-fit results for $N_{\rm ev} = 71$ events, $N_{\rm bin} = 15$ bins per quantile (30 histogram entries total), bin edges set to the observed quantile range $\pm 10\%$ margin. $N_{\rm params}$ is the number of free parameters; $N_{\rm eff}$ is the effective injection sample size after re-weighting; $\chi^2_{\rm min} = \chi^2(\hat{\boldsymbol{\theta}})$ is the test statistic at the best-fit $\hat{\boldsymbol{\theta}}$; p is the Monte Carlo calibrated p-value ($N_{\rm mock} = 500$).

Model	$N_{\rm params}$	$N_{\rm eff}$	$\chi^2_{\rm min}$	p-value	$\boldsymbol{\theta}_{\rm BF}$
Isotropic	0	8728	41.02	0.074	—
Skew-normal	3	4696	9.21	0.548	a=13.44, ξ=−0.069, ω=0.161
LVK Default	5	3296	20.05	0.146	μ_χ=0.091, σ_χ=0.318, μ_t=0.752, σ_t=0.884, ζ=0.382
Truncated Gaussian	2	5006	15.69	0.230	μ=0.042, σ=0.103
Roulet+ 2021	3	4444	13.72	0.258	ζ_+=0.45, ζ_-=0.00, σ=0.23

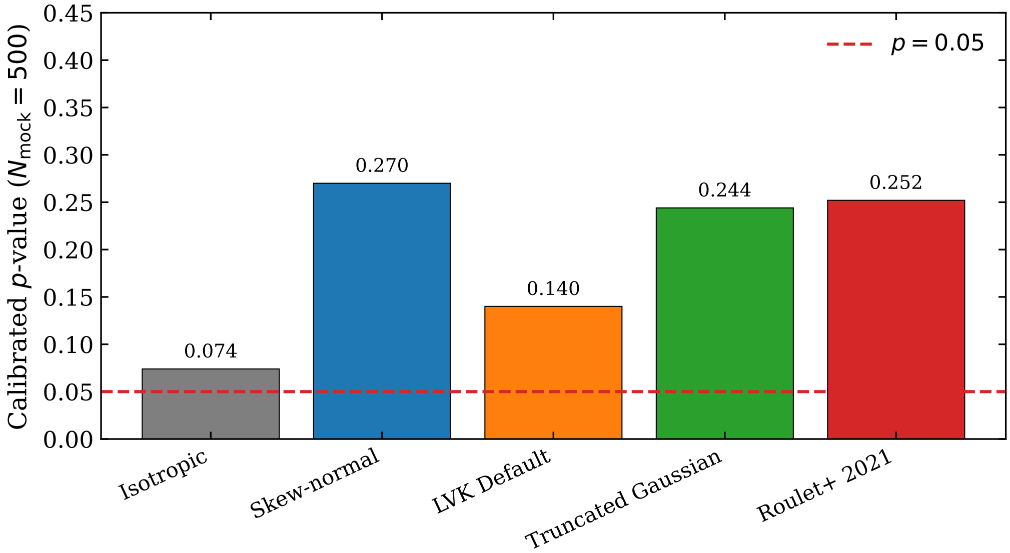

Figure 2: Summary of calibrated p-values for all five $\chi_{\rm eff}$ population models.

3 Binning-free diagnostic: CDF comparison

The χ^2 test statistic depends on the choice of histogram binning. We find that p-values change appreciably when the number of bins is varied (e.g. from $N_{\rm bin} = 10$ to 15), because the covariance matrix is sensitive to the number of empty or near-empty bins.

As a complementary, binning-free diagnostic, we compare the empirical cumulative distribution functions (CDFs) of the observed and predicted $\chi_{\rm eff}$ quantiles. For a given $\boldsymbol{\theta}$, the model CDF is the importance-weighted empirical CDF of the injection quantiles:

$$F_{\rm model}(x \mid \boldsymbol{\theta}) = \frac{\sum_{k:\, q_k \le x} w_k(\boldsymbol{\theta})}{\sum_k w_k(\boldsymbol{\theta})}. \tag{10}$$

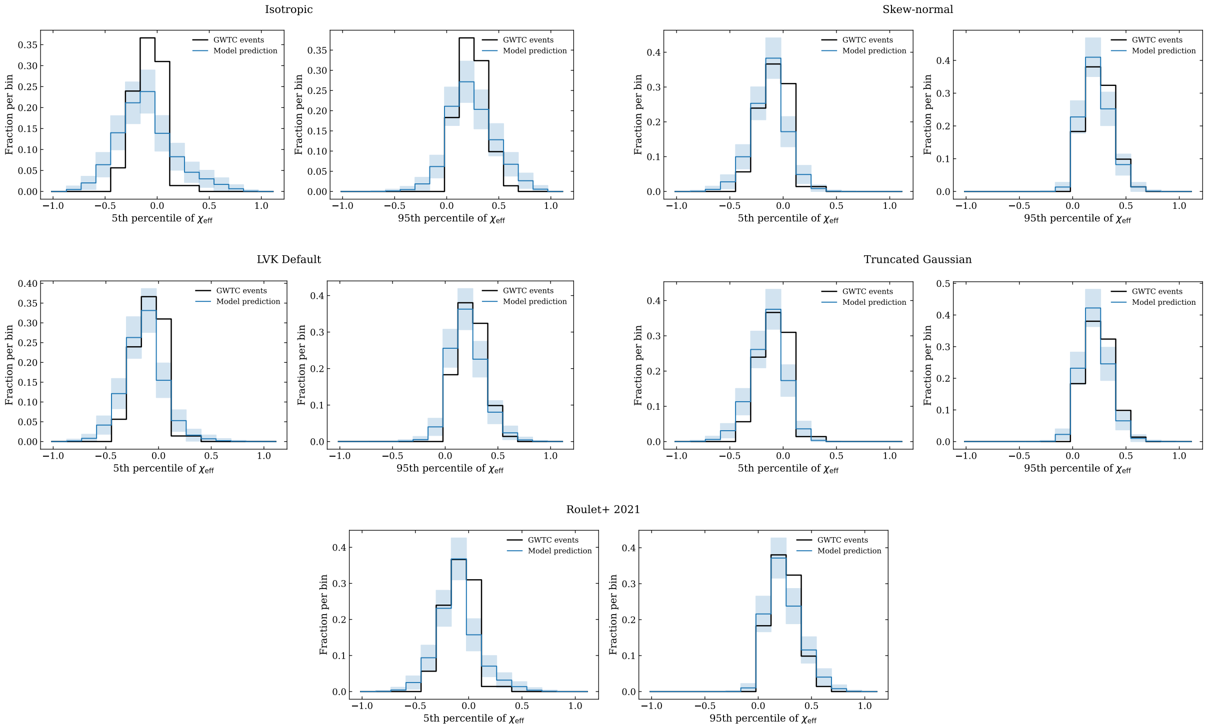

Figure 3: Histogram overlays of observed (data) vs. predicted (model) quantile distributions for all five models. The model prediction and 1σ bootstrap band are evaluated at $\hat{\boldsymbol{\theta}}$.

The test statistic is the two-sample Kolmogorov–Smirnov distance between the observed and model CDFs, summed over both quantiles:

$$D(\boldsymbol{\theta}) = \sup_x \left| F^5_{\rm obs}(x) - F^5_{\rm model}(x) \right| + \sup_x \left| F^{95}_{\rm obs}(x) - F^{95}_{\rm model}(x) \right|. \tag{11}$$

This is evaluated without any binning and requires only sorting (~ 1 ms per evaluation, compared to ~ 100 ms for the bootstrap-based χ^2).

The calibrated p-value follows the same logic as in Section 1: minimise $D(\boldsymbol{\theta})$ for the real catalogue (Nelder–Mead, 3 restarts), generate mock catalogues from $\boldsymbol{\theta}_{\rm BF}$, let each mock minimise $D(\boldsymbol{\theta})$, and compute $p = N_{\rm mock}^{-1} \sum_j \mathbb{1}[D_{{\rm min},j} \geq D_{\rm min,obs}]$.

Figures 5 and 6 show the empirical CDF comparison for the 5th and 95th percentiles, respectively. The model CDF and bootstrap band are evaluated at $\hat{\boldsymbol{\theta}}_{\rm KS}$ (the KS-minimised parameters from Table 3).

Parameter shifts under minimisation. An important diagnostic is whether $\hat{\boldsymbol{\theta}}$ moves significantly from $\boldsymbol{\theta}_{\rm BF}$ during minimisation. Table 2 compares the two for the χ^2 and KS statistics. The χ^2 optimiser moves modestly from $\boldsymbol{\theta}_{\rm BF}$ for most models. The KS optimiser moves further, particularly for the LVK Default model where the tilt parameters shift substantially ($\sigma_t : 0.88 \to 0.13$, $\zeta : 0.38 \to 0.77$). Large shifts indicate that the hierarchical best-fit $\boldsymbol{\theta}_{\rm BF}$ does not minimise the GoF statistic, i.e. the population model at its published parameters is not the best-fitting member of the model family for this observable.

Table 3 summarises the KS-based results.

4 Discussion

The calibrated p-values quantify how well each population model reproduces the observed distribution of $\chi_{\rm eff}$ posterior quantiles. Models with $p \lesssim 0.05$ would be disfavoured at the 95%

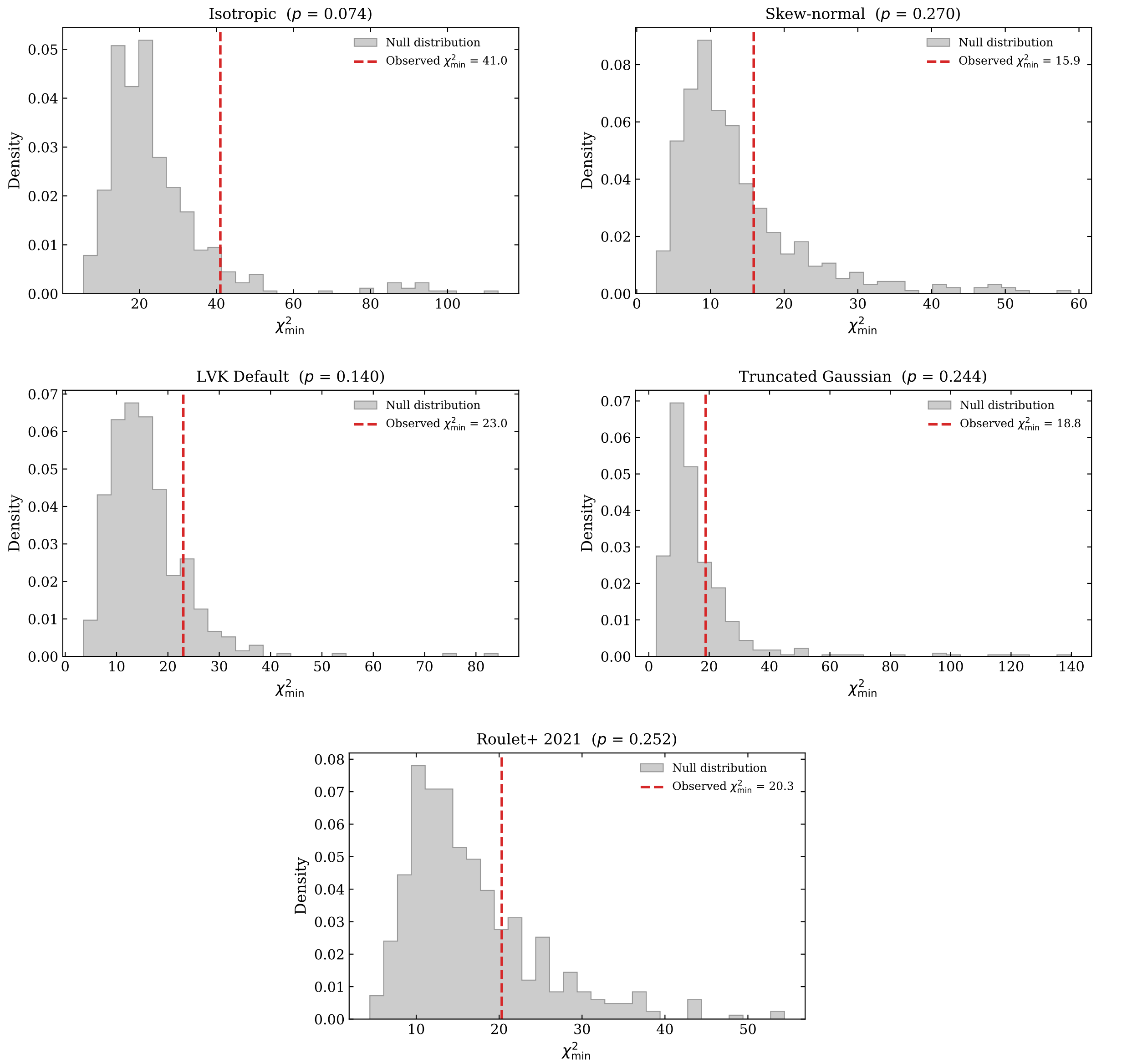

Figure 4: Null distributions of $\chi^2_{\rm min}$ from $N_{\rm mock} = 500$ mock catalogues for all five models. The vertical dashed line marks the observed $\chi^2_{\rm min}$.

Table 2: Best-fit parameters $\hat{\boldsymbol{\theta}}$ from χ^2 and KS minimisation, compared to $\boldsymbol{\theta}_{\rm BF}$ from hierarchical inference. Only models with free parameters are shown. †Nelder–Mead exited the physical parameter bounds ($\zeta_+ > 1$); result invalid and being recomputed with bounded optimisation.

Model	$\boldsymbol{\theta}_{\rm BF}$	$\hat{\boldsymbol{\theta}}_{\chi^2}$	$\hat{\boldsymbol{\theta}}_{\rm KS}$
Skew-normal	$(13.44, -0.069, 0.161)$	$(16.40, -0.041, 0.174)$	$(8.89, 0.011, 0.123)$
LVK Default	$(0.091, 0.318, 0.752, 0.884, 0.382)$	$(0.091, 0.318, 0.752, 0.884, 0.382)$	(0.[illegible]10, 0.277, 0.457, 0.131, 0.774)
Trunc. Gauss.	$(0.042, 0.103)$	$(0.045, 0.103)$	$(0.105, 0.080)$
Roulet+	$(0.45, 0.00, 0.23)$	$(0.50, 0.00, 0.18)$	invalid†

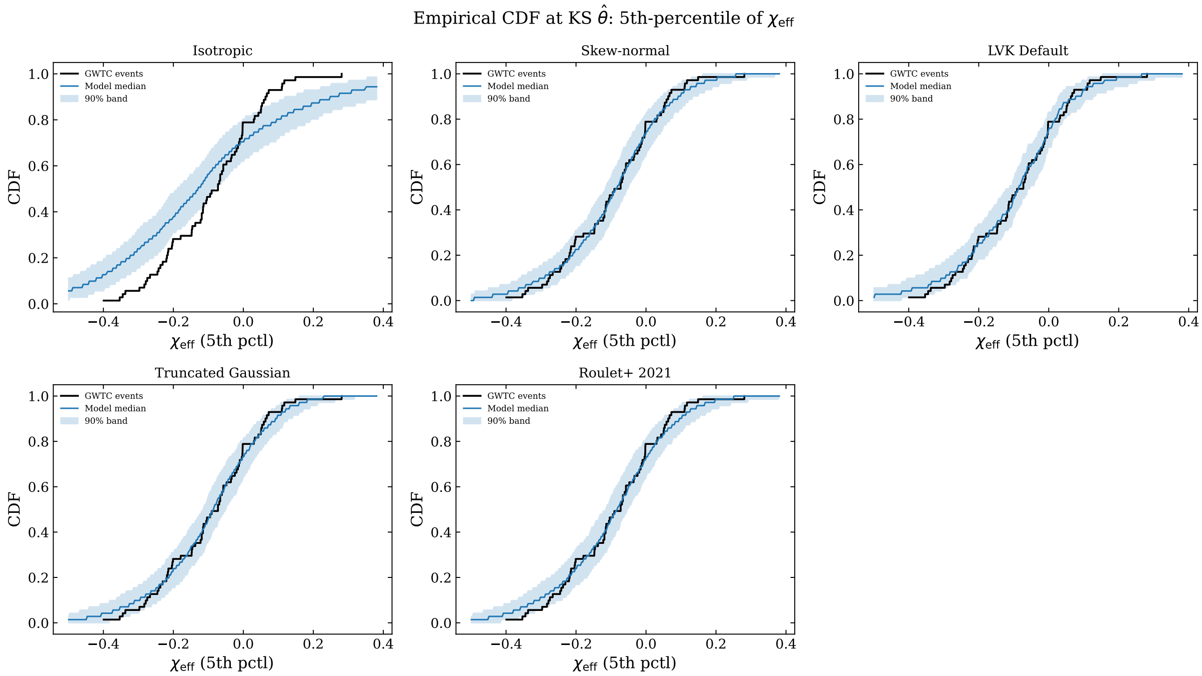

Figure 5: Empirical CDF comparison for the 5th percentile of $\chi_{\rm eff}$. Black step function: observed events (71 steps). Blue curve: median model CDF at $\hat{\boldsymbol{\theta}}_{\rm KS}$. Blue band: 90% envelope from 500 bootstrap draws of 71 events from the model, showing the expected scatter from finite catalogue size. Where the black curve exits the band, the model is in tension with the data.

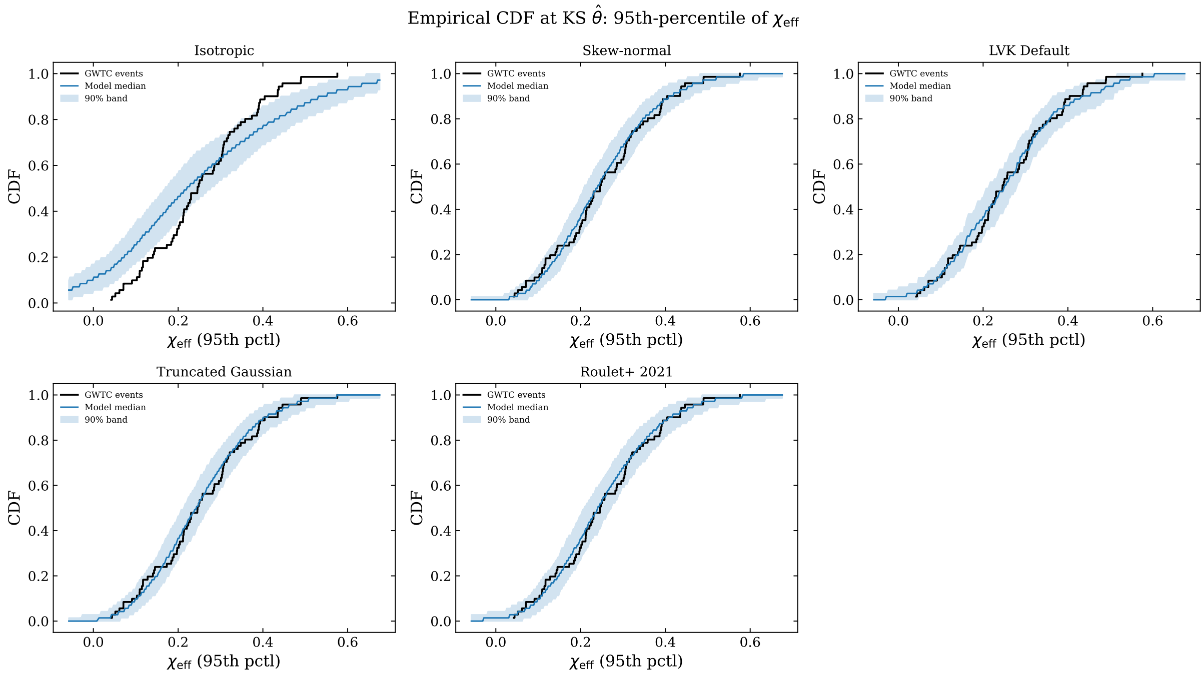

Figure 6: Same as Figure 5 but for the 95th percentile of $\chi_{\rm eff}$.

Table 3: Binning-free goodness-of-fit results using the KS test statistic (11). $D_{\min}$ is the minimised KS distance; p is the calibrated p-value ($N_{\text{mock}} = 500$). The isotropic model is rejected at the 5% level ($p = 0.024$).

Model	$D_{\min}$	p-value (χ^2)	p-value (KS)
Isotropic	0.330	0.074	**0.024**
Skew-normal	0.113	0.548	0.706
LVK Default	0.097	0.146	0.802
Truncated Gaussian	0.113	0.230	0.742
Roulet+ 2021	0.112	0.258	0.640

level.

Key caveats:

- The χ^2 test is sensitive to the choice of binning; the CDF-based KS test provides a binning-free cross-check.
- The effective sample size N_{eff} of the re-weighted injection set varies across models, affecting the reliability of the predicted distributions.
- For events with weak χ_{eff} constraints, the PE posterior quantiles are dominated by the PE prior rather than the signal, reducing the discriminating power of the test.
- Free-parameter models benefit from the flexibility of fitting; the mock calibration accounts for this look-elsewhere effect.